

Unseen Seminar Questions 3: Answers

Probability and Statistical Inference

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Reference formulas for Question 1.

Memorising these gallery formulas is non-examinable. You may use:

Distribution	Mean	Variance
$B \sim \text{Binomial}(m, p)$	mp	$mp(1 - p)$
$G \sim \text{Gamma}(a, \lambda)$, shape–rate	a/λ	a/λ^2
$P \sim \text{Beta}(a, b)$	$a/(a + b)$	
$\log Y \sim N(\mu, \sigma^2)$	$e^{\mu + \sigma^2/2}$	

1. **Choosing a probability model.** For each setting, name a suitable distribution and calculate the requested quantity.
 - (a) Twelve patients recover independently, each with probability 0.7. Model the number who recover and find its mean and variance.
 - (b) Use the model for counts and waiting times from Lecture 3, with incidents arriving at a constant rate of 2 per day and independent counts in disjoint intervals. Model the waiting time to the third incident and find its mean and variance.
 - (c) A random proportion P is modeled flexibly on $(0, 1)$ with parameters $a = 2$ and $b = 5$. Name the distribution and find its mean.
 - (d) A positive income variable satisfies $\log Y \sim N(10, 0.25)$. Name the distribution of Y and give its median and mean.

Solution.

- (a) If R is the number who recover, then

$$R \sim \text{Binomial}(12, 0.7), \quad \mathbb{E}[R] = 12(0.7) = 8.4, \quad \text{Var}(R) = 12(0.7)(0.3) = 2.52.$$

- (b) If T_3 is the waiting time in days, then in shape–rate notation

$$T_3 \sim \text{Gamma}(3, 2), \quad \mathbb{E}[T_3] = \frac{3}{2}, \quad \text{Var}(T_3) = \frac{3}{2^2} = \frac{3}{4}.$$

- (c) The model is $P \sim \text{Beta}(2, 5)$. Its mean is

$$\mathbb{E}[P] = \frac{2}{2 + 5} = \frac{2}{7}.$$

- (d) The variable Y is log-normal. Since the normal mean and variance of $\log Y$ are $\mu = 10$ and $\sigma^2 = 0.25$,

$$\text{Median}(Y) = e^{10}, \quad \mathbb{E}[Y] = e^{\mu + \sigma^2/2} = e^{10.125}.$$

The mean exceeds the median because of the right skew.

2. **Uncorrelated but dependent.** Let X take values $-2, 0, 2$ with probabilities $1/4, 1/2, 1/4$, respectively, and define $Y = X^2$.

- (a) Calculate $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$ and $\text{Cov}(X, Y)$.
- (b) Show directly that X and Y are not independent.
- (c) Why does this example not conflict with the result that uncorrelated jointly normal variables are independent?

Solution.

- (a) Symmetry gives

$$\mathbb{E}[X] = (-2)\frac{1}{4} + 0\frac{1}{2} + 2\frac{1}{4} = 0.$$

Since $Y = 4$ when $X = \pm 2$ and $Y = 0$ when $X = 0$,

$$\mathbb{E}[Y] = 4\left(\frac{1}{4} + \frac{1}{4}\right) = 2.$$

Also $XY = X^3$, so

$$\mathbb{E}[XY] = (-8)\frac{1}{4} + 0\frac{1}{2} + 8\frac{1}{4} = 0.$$

It follows that

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

- (b) The variables cannot be independent because observing X determines Y exactly. For example,

$$\mathbb{P}(Y = 0 \mid X = 0) = 1, \quad \mathbb{P}(Y = 0) = \frac{1}{2}.$$

These probabilities would agree under independence.

- (c) The pair is discrete and satisfies the nonlinear deterministic relation $Y = X^2$; it is not jointly normal. The implication from zero covariance to independence is a special property of jointly normal variables, so it does not apply here.

3. **Tail behavior and moment existence.** Three symmetric densities have the following forms for $|x| \geq 1$:

$$f_1(x) = c_1|x|^{-5/2}, \quad f_2(x) = c_2|x|^{-3}, \quad f_3(x) = c_3|x|^{-4},$$

where the constants are positive. For each distribution, decide whether its mean exists and whether its variance is finite. In the boundary case for the second moment, explain why an integrand approaching zero does not by itself imply a finite integral.

Solution. If a density has tails proportional to $|x|^{-q}$, multiply it by $|x|^r$ to form the integrand for the absolute moment of order r . In the positive tail, this integrand is proportional to

$$x^r x^{-q} = x^{r-q}.$$

Its tail integral is finite exactly when $r - q < -1$, equivalently $q > r + 1$. All three exponents are greater than 2, so their first absolute moments are finite. Their means therefore exist and, by symmetry, equal zero.

For the second moment we need $q > 3$. Thus f_3 has a finite second moment and finite variance, while f_1 and f_2 have infinite second moments. For f_2 the second-moment integrand behaves as

$$x^2 f_2(x) = \frac{c_2}{x}.$$

This tends to zero, but $\int_1^M x^{-1} dx = \log M$ diverges as $M \rightarrow \infty$. Pointwise convergence of an integrand to zero is therefore not enough to give a finite integral.

4. **Read the covariance matrix.** Conditional on the regressors X , consider three possible covariance matrices for $\varepsilon = (\varepsilon_1, \varepsilon_2, \varepsilon_3)'$:

$$\Omega_A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \Omega_B = \begin{pmatrix} 2 & 0.8 & 0.2 \\ 0.8 & 2 & 0.8 \\ 0.2 & 0.8 & 2 \end{pmatrix}, \quad \Omega_C = \begin{pmatrix} 1 & 0.6 & 0 \\ 0.6 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

For interpreting Ω_B , suppose the observations are ordered in time. For Ω_C , suppose observations 1 and 2 belong to one sampling cluster and observation 3 belongs to another. These three-observation matrices illustrate covariance patterns.

- (a) Describe the variance and dependence pattern in each matrix.
 (b) Which matrix has the form $\sigma^2 I_3$? With X of full column rank, show how the general OLS covariance formula

$$(X'X)^{-1}X'\Omega X(X'X)^{-1}$$

simplifies for this matrix. Explain why the same simplification is not justified by either of the other covariance patterns.

- (c) Calculate $\text{Var}(\varepsilon_1 - \varepsilon_2 \mid X)$ under each matrix.

Solution.

- (a) In Ω_A , every variance equals 2 and every cross-observation covariance is zero: the errors are homoscedastic and uncorrelated. In Ω_B , every variance still equals 2, but observations are positively correlated; the covariance is larger for the adjacent pairs (1, 2) and (2, 3) than for (1, 3), a serial-correlation pattern. In Ω_C , variances differ and the first two observations are correlated while the third is uncorrelated with them. This is a heteroscedastic, block or clustered pattern with observations 1 and 2 in one cluster.
 (b) Only $\Omega_A = 2I_3$ has equal diagonal entries and zero off-diagonal entries. Substituting it gives

$$(X'X)^{-1}X'(2I_3)X(X'X)^{-1} = 2(X'X)^{-1}.$$

The cancellation uses $(X'X)^{-1}X'X = I$. Ω_B has nonzero off-diagonal entries, and Ω_C has both unequal variances and a nonzero off-diagonal covariance. Neither has the required form $\sigma^2 I_3$, so their patterns do not justify replacing the general formula by $\sigma^2(X'X)^{-1}$.

- (c) For any covariance matrix,

$$\text{Var}(\varepsilon_1 - \varepsilon_2 \mid X) = \text{Var}(\varepsilon_1 \mid X) + \text{Var}(\varepsilon_2 \mid X) - 2\text{Cov}(\varepsilon_1, \varepsilon_2 \mid X).$$

It is therefore

matrix	$\text{Var}(\varepsilon_1 - \varepsilon_2 \mid X)$
Ω_A	$2 + 2 - 2(0) = 4$
Ω_B	$2 + 2 - 2(0.8) = 2.4$
Ω_C	$1 + 4 - 2(0.6) = 3.8$

Positive covariance reduces the variance of the difference.